

The Complete Machine Learning Curriculum

A Comprehensive Guide from Fundamentals to Generative AI

2026 Edition

Chapter 1: Foundations & Supervised Learning

Topics in this chapter focus on mapping inputs to known outputs using labeled datasets.

Linear Models:

Linear Regression, Logistic Regression, Polynomial Regression, GLMs.

Tree-Based Models:

Decision Trees, Random Forests, Gradient Boosting (XGBoost, LightGBM, CatBoost).

Distance & Support Vectors:

K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Kernel Trick.

Evaluation Metrics:

MSE, MAE, R-Squared, Accuracy, Precision-Recall, F1-Score, ROC-AUC.

Chapter 2: Unsupervised Learning & Preprocessing

This chapter explores finding hidden patterns in data without explicit labels.

Clustering:

K-Means, K-Medoids, Hierarchical Clustering (Agglomerative), DBSCAN, Gaussian Mixture Models (GMM).

Dimensionality Reduction:

Principal Component Analysis (PCA), t-SNE, UMAP, Linear Discriminant Analysis (LDA).

Association & Anomaly:

Apriori Algorithm, Eclat, Isolation Forests, Local Outlier Factor (LOF).

Chapter 3: Deep Learning (Neural Networks)

Topics include the architecture and optimization of biologically inspired models.

Architecture Fundamentals:

Perceptrons, Multi-Layer Perceptrons (MLP), Activation Functions (ReLU, Sigmoid, Tanh, Softmax).

Optimization:

Backpropagation, Gradient Descent (SGD, Adam, RMSProp), Weight Initialization (Xavier, He).

Regularization:

Dropout, Batch Normalization, L1/L2 Regularization, Early Stopping.

Chapter 4: Natural Language Processing (NLP)

This chapter covers techniques to process, analyze, and generate human language.

Text Preprocessing:

Tokenization, Stopword Removal, Stemming, Lemmatization, Part-of-Speech (POS) Tagging.

Vectorization:

Bag of Words, TF-IDF, Word Embeddings (Word2Vec, GloVe, FastText).

Sequence Modeling:

RNNs, LSTMs, GRUs, Bidirectional RNNs.

Modern Architectures:

Attention Mechanisms, Transformers (Encoder-only, Decoder-only), BERT, RoBERTa.

Chapter 5: Computer Vision (CV)

Topics focused on enabling machines to see and interpret visual data.

Core CNNs:

Convolutional Layers, Pooling (Max/Average), Stride, Padding, Fully Connected Layers.

Famous Architectures:

AlexNet, VGG, ResNet (Residual Connections), Inception, MobileNet, EfficientNet.

Tasks:

Object Detection (YOLO, R-CNN), Image Segmentation (U-Net, Mask R-CNN), Face Recognition.

Chapter 6: Generative AI & LLMs

The cutting edge of AI focusing on the creation of new content.

Generative Models:

Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Diffusion Models.

Large Language Models (LLMs):

GPT-3/4, T5, Llama, PaLM. Fine-tuning (LoRA, QLoRA), Prompt Engineering.

Advanced Frameworks:

Retrieval Augmented Generation (RAG), LangChain, LangGraph (Agentic Workflows), Vector Databases (Pinecone, Milvus).

Chapter 7: Reinforcement Learning

Learning through trial and error to maximize rewards.

Concepts:

Markov Decision Processes (MDP), Bellman Equation, Rewards, Policies.

Algorithms:

Q-Learning, SARSA, Deep Q-Networks (DQN), Policy Gradients, PPO, Actor-Critic.